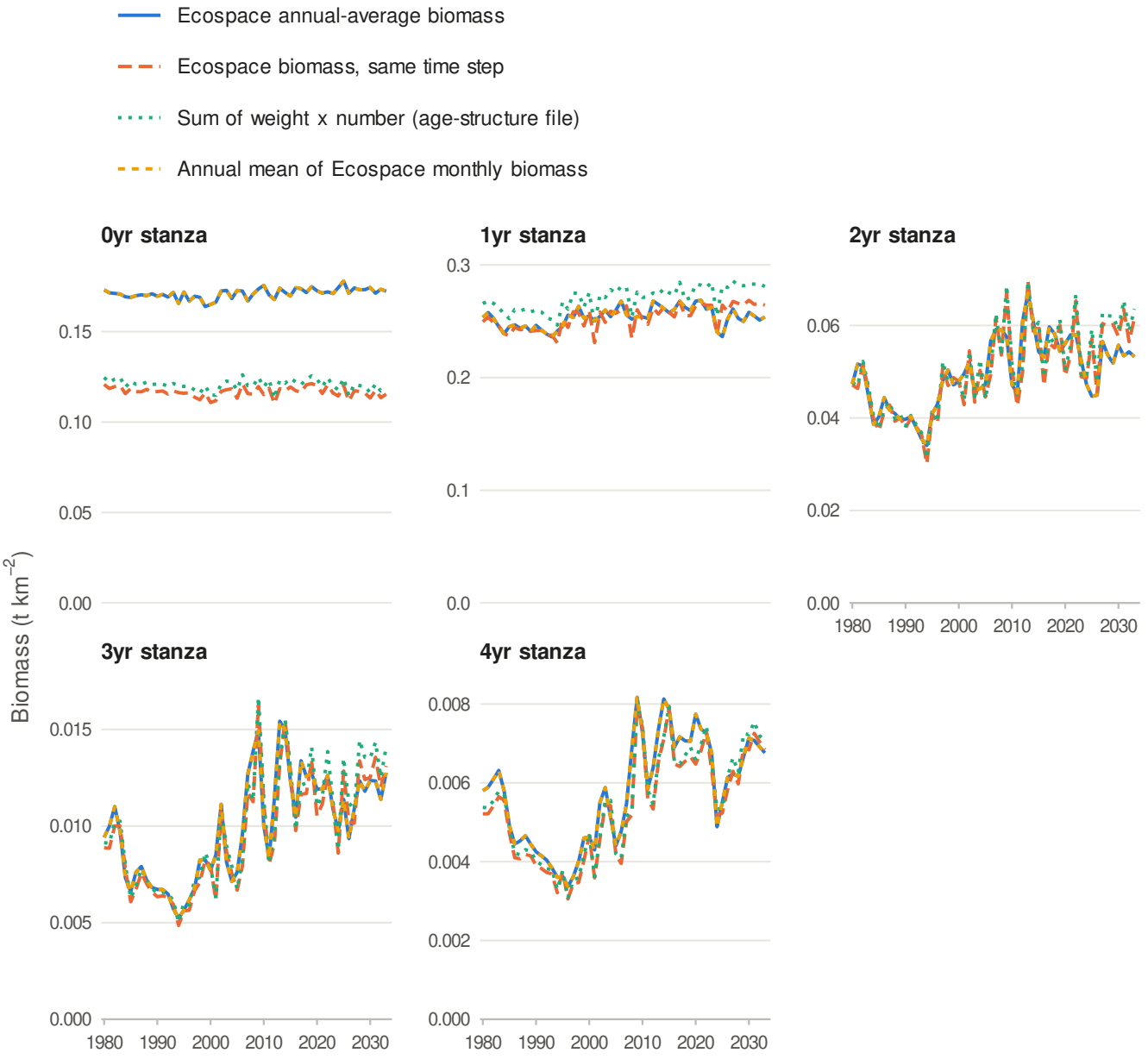
Menhaden age-structure output vs. Ecospace stanza biomass

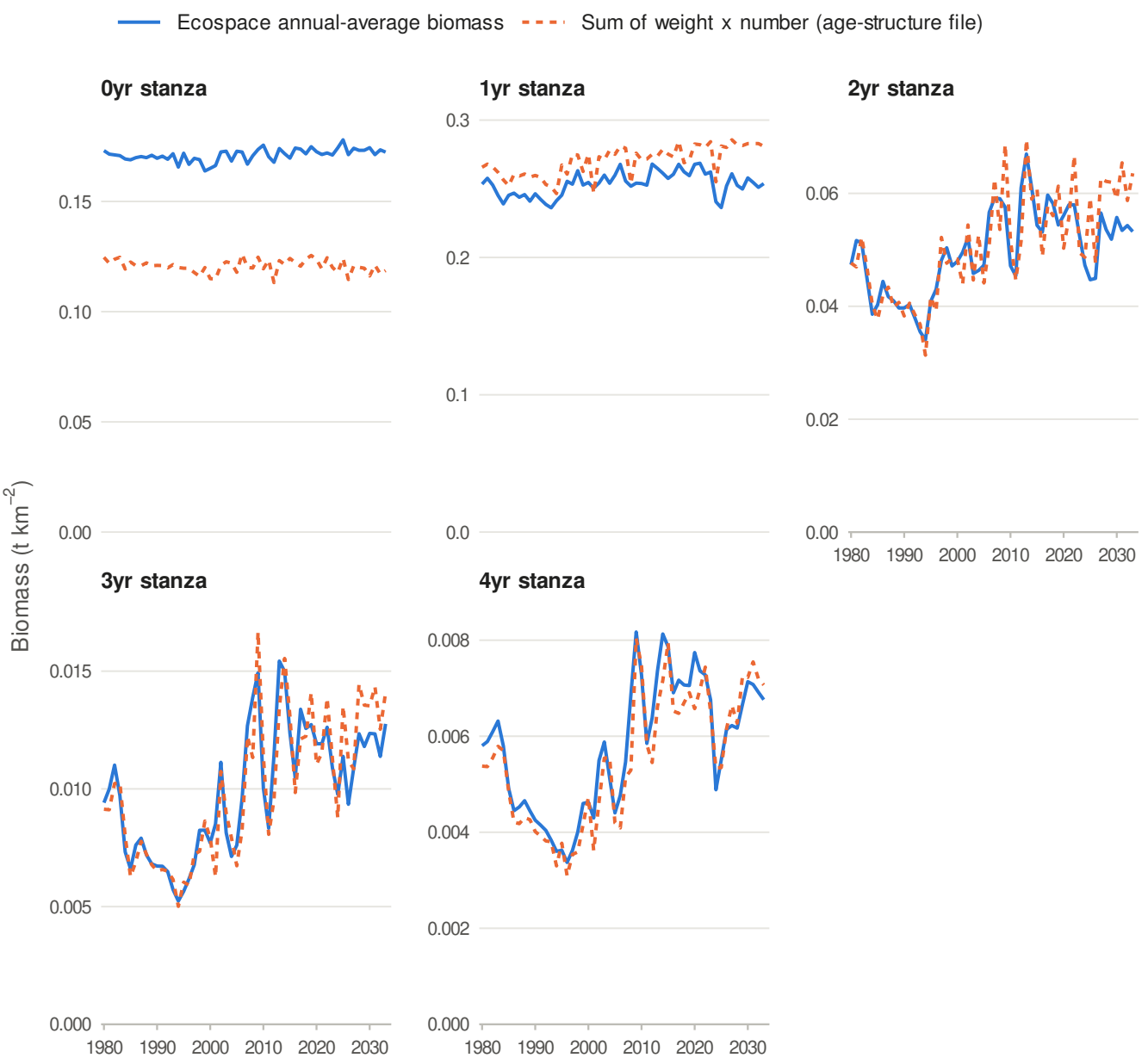
Ecospace biomass outputs vs. biomass rebuilt from the January age structure



Monthly-biomass annual mean averages Ecospace output; it is not an independent numbers x weight reconstruction.

Menhaden age-structure output vs. Ecospace stanza biomass

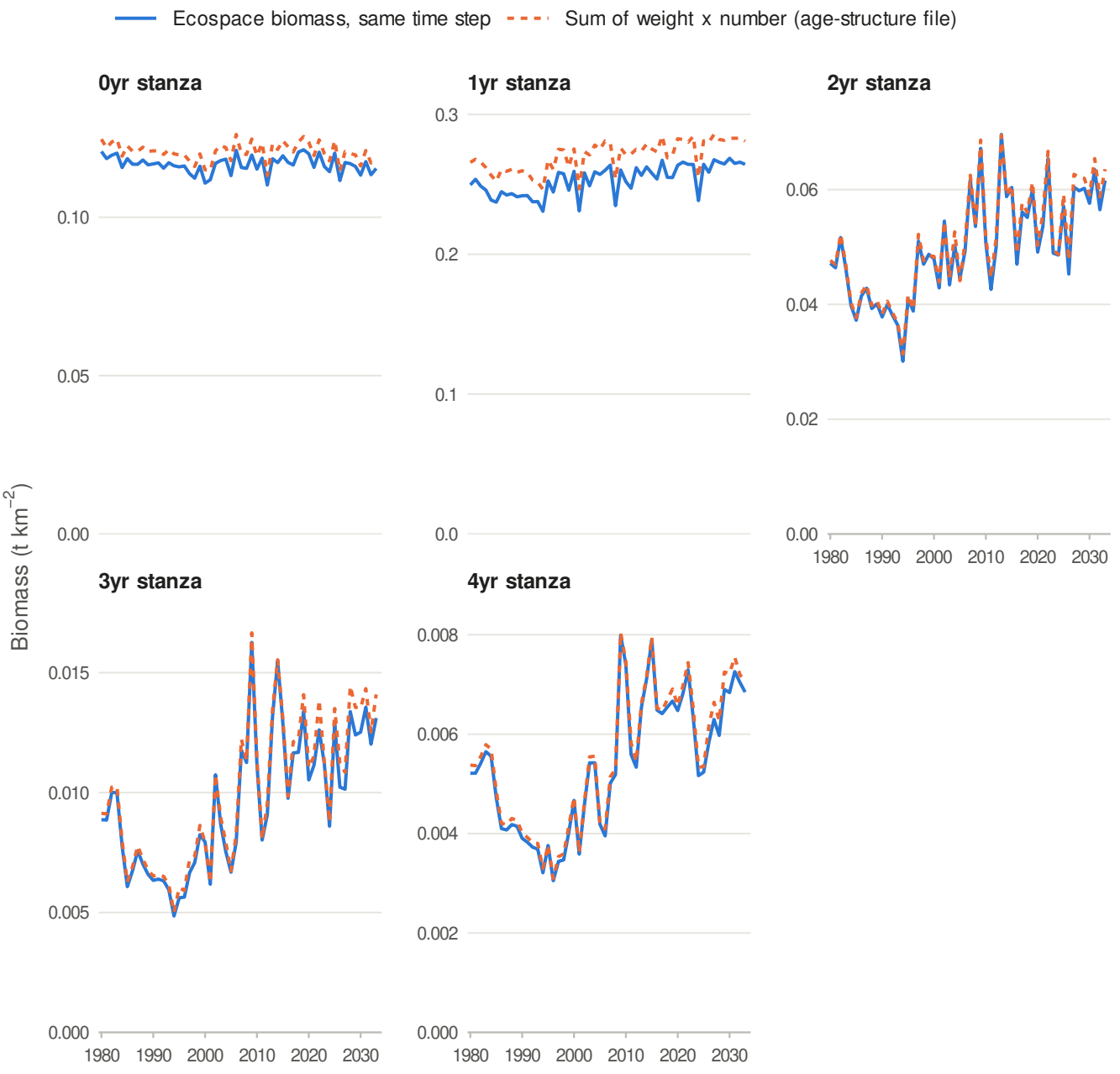
(a) Annual average vs. January snapshot of the age-structure file



Age-0 differs by ~1.4x because the January snapshot precedes the seasonal recruitment pulse.

Menhaden age-structure output vs. Ecospace stanza biomass

(b) Month-for-month: both taken from the same January time step



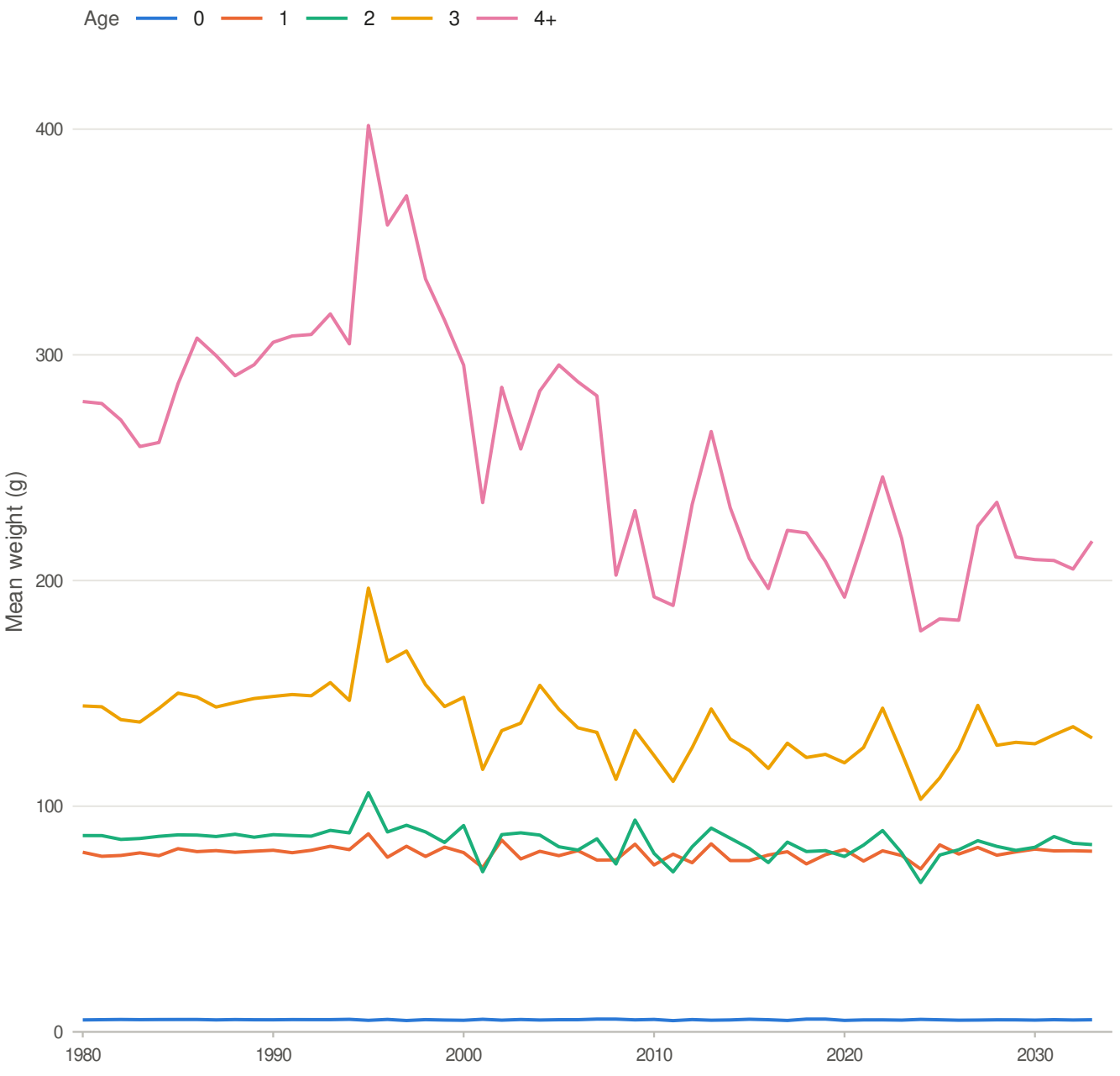
Menhaden mortality by age, 1980-2032

Z from cohort decline between January snapshots; F from catch / biomass, weighted by stanza exposure



Menhaden weight-at-age

Number-weighted mean body weight on 1 January (W-infinity = 696 g)



Menhaden natural mortality: two independent estimates

M2 from the Ecospace consumption matrix plus Ecopath M0, vs. M derived from cohort decline



Predation mortality on menhaden by predator

M2 = predator consumption of each stanza / stanza biomass (Ecospace annual averages)

